In [3]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [16]:
```
iris = sns.load_dataset('iris')
```

In [18]:
```
print("First 5 rows of dataset:")
print(iris.head())
```

```
First 5 rows of dataset:
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

In [20]:
```
print("\nDataset Info:")
print(iris.info())
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
```

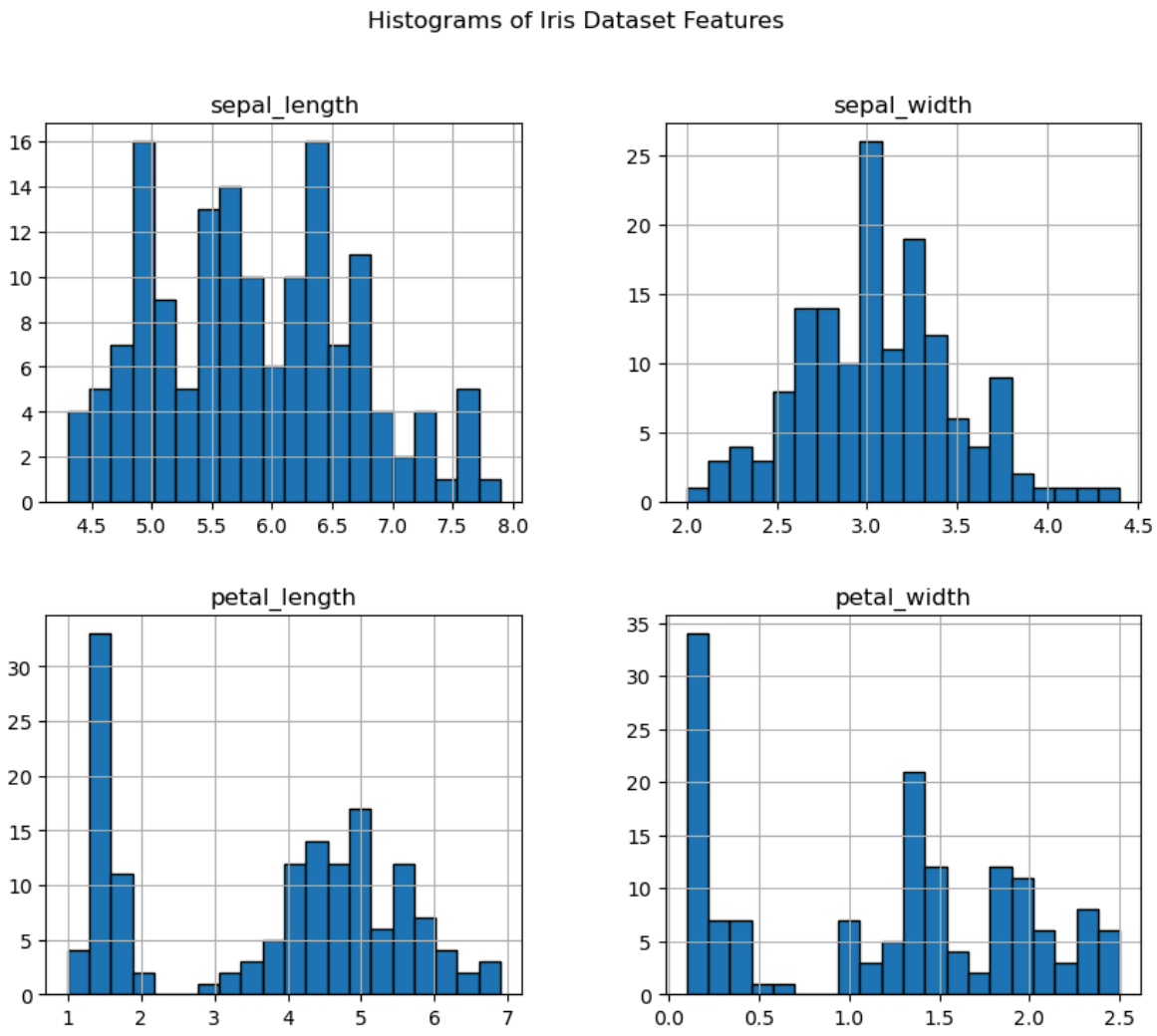
In [22]:
```
print("\nFeature Types:")
for column in iris.columns:
    if iris[column].dtype == 'object':
        print(f"{column} : Nominal (Categorical)")
    else:
        print(f"{column} : Numeric")
```

```
Feature Types:
sepal_length : Numeric
sepal_width : Numeric
petal_length : Numeric
petal_width : Numeric
species : Nominal (Categorical)
```

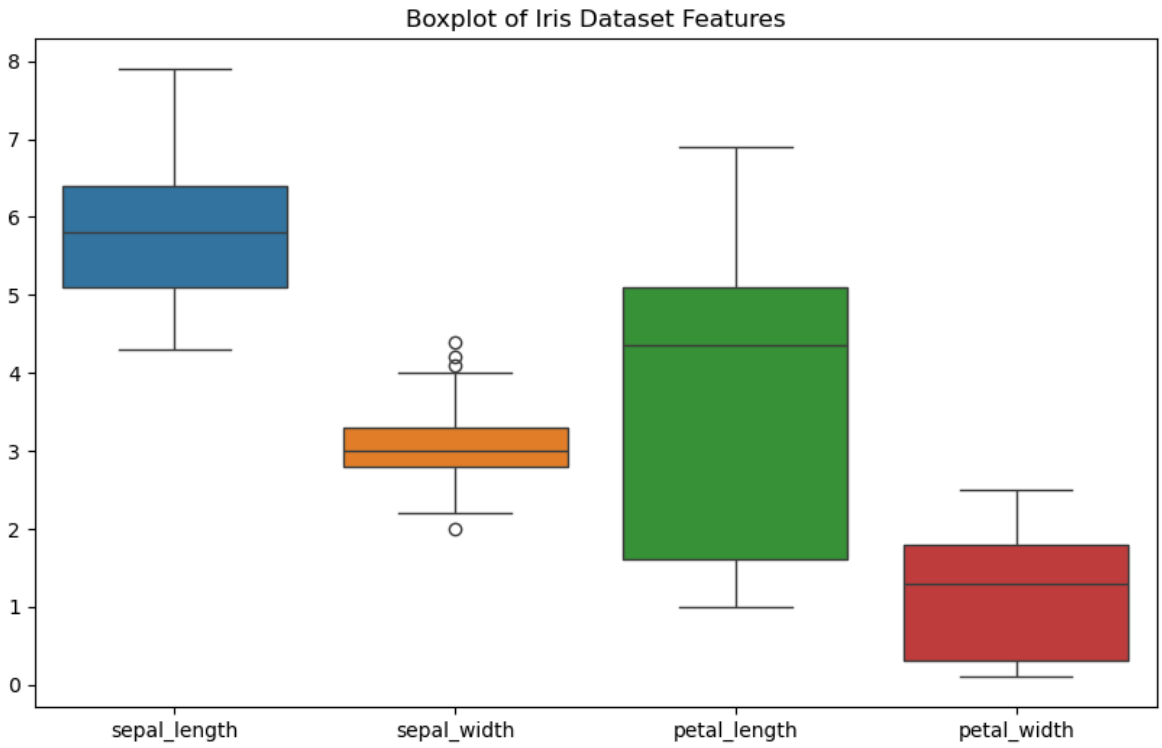
In [24]:
```
print("\nStatistical Summary:")
print(iris.describe())
```

```
Statistical Summary:
       sepal_length  sepal_width  petal_length  petal_width
count    150.000000   150.000000    150.000000   150.000000
mean       5.843333     3.057333      3.758000     1.199333
std        0.828066     0.435866      1.765298     0.762238
min        4.300000     2.000000      1.000000     0.100000
25%        5.100000     2.800000      1.600000     0.300000
50%        5.800000     3.000000      4.350000     1.300000
75%        6.400000     3.300000      5.100000     1.800000
max        7.900000     4.400000      6.900000     2.500000
```

In [26]:
```
iris.hist(figsize=(10, 8), bins=20, edgecolor='black')
plt.suptitle("Histograms of Iris Dataset Features")
plt.show()
```

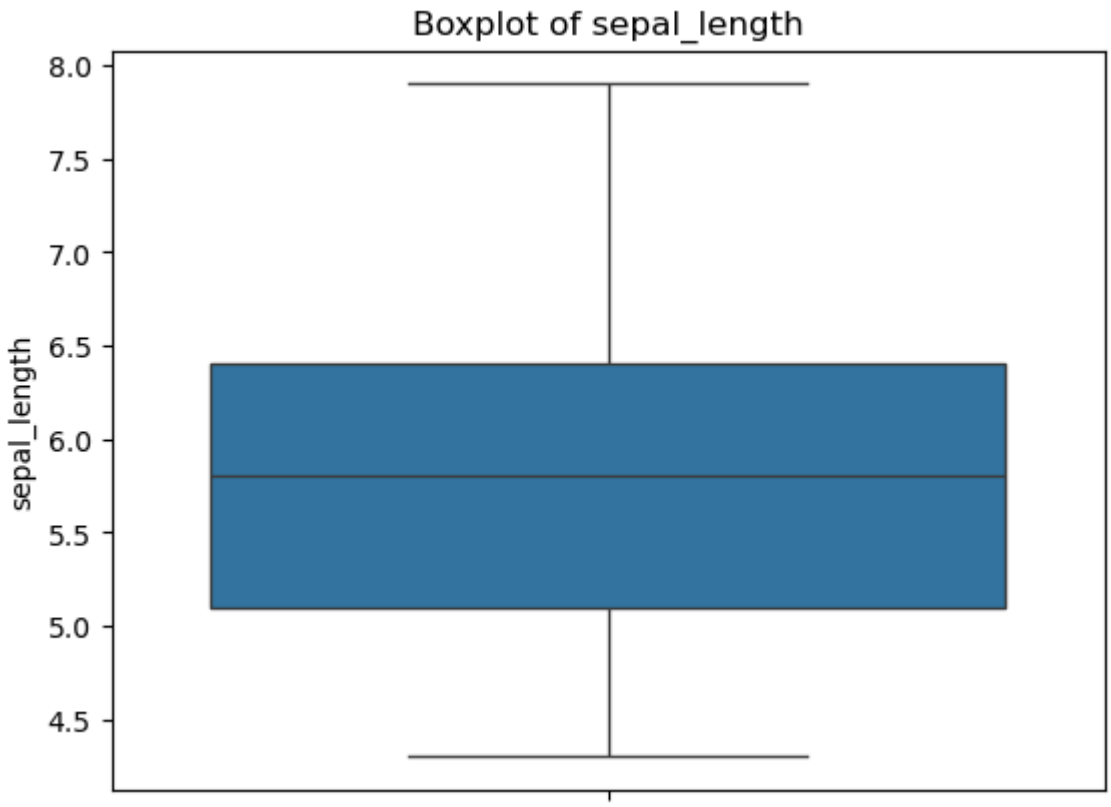


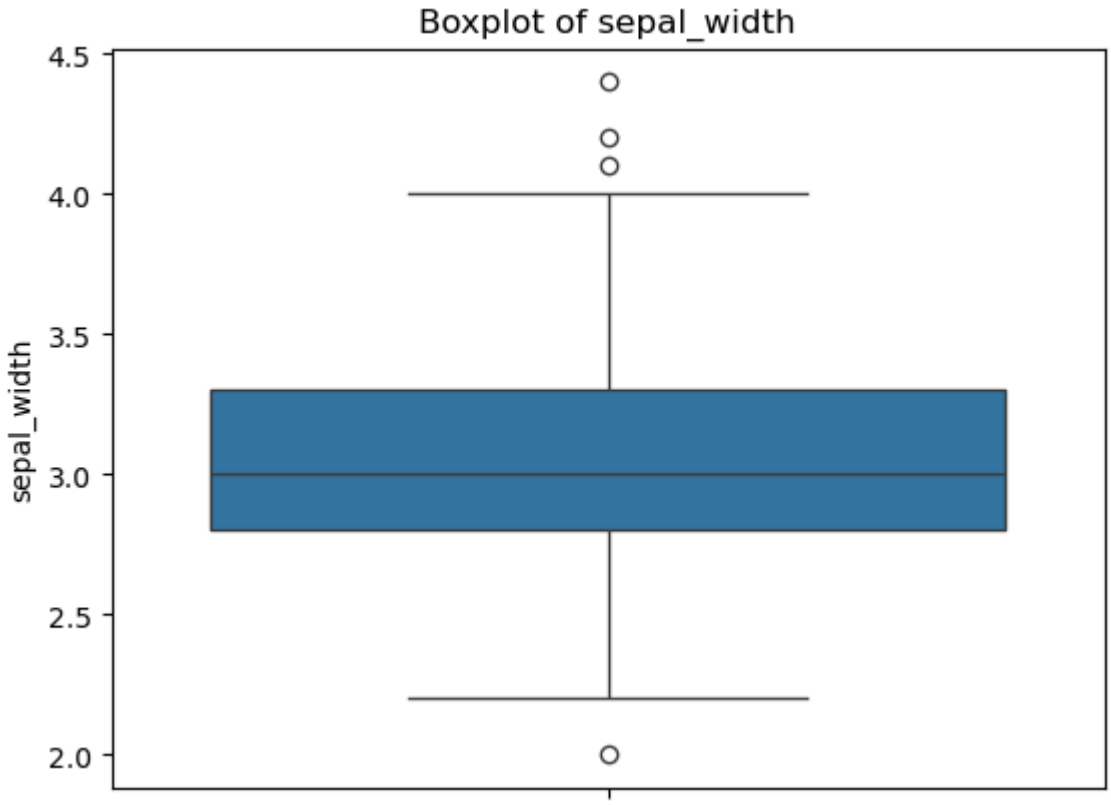
In [28]:
```
plt.figure(figsize=(10, 6))
sns.boxplot(data=iris)
plt.title("Boxplot of Iris Dataset Features")
plt.show()
```

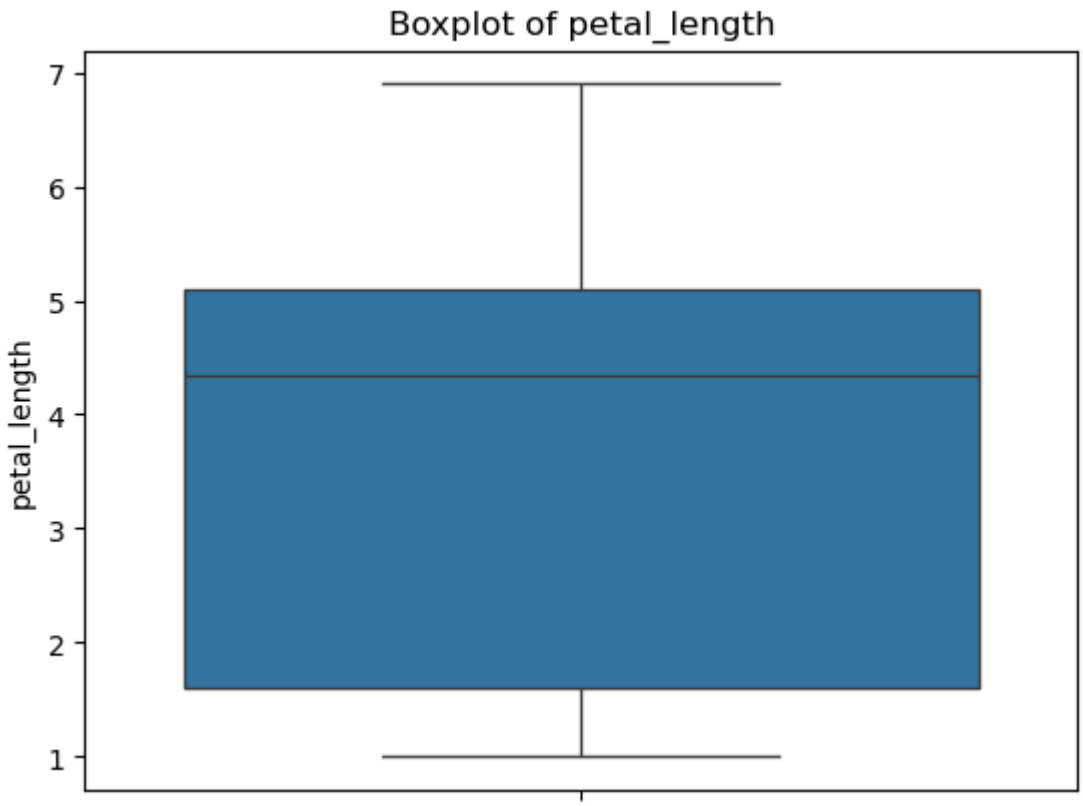


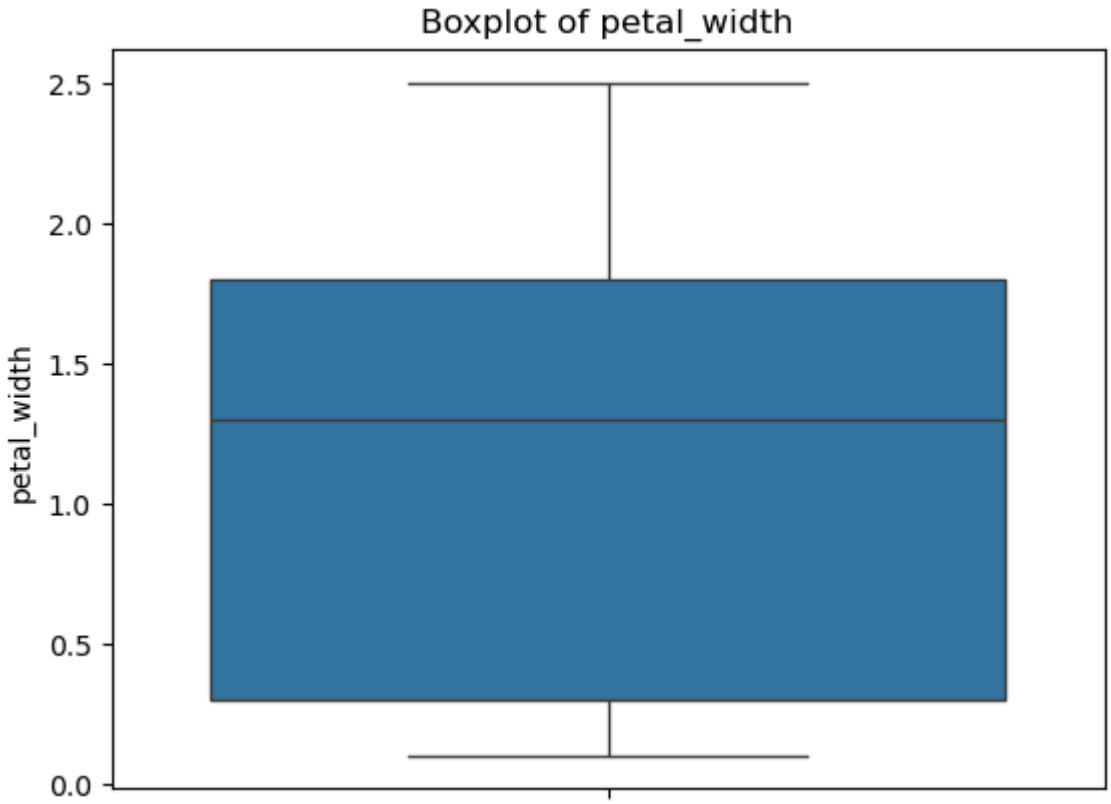
In [30]:
```
features = iris.columns[:-1]

for feature in features:
    plt.figure()
    sns.boxplot(y=iris[feature])
    plt.title(f"Boxplot of {feature}")
    plt.show()
```




Boxplot of sepal_width
sepal_width
4.5
4.0
3.5
3.0
2.5
2.0


Boxplot of petal_length
petal_length
7
6
5
4
3
2
1



In [32]:
```
# Step 10: Identify outliers using IQR method
print("\nOutlier Detection using IQR:\n")

for feature in features:
    Q1 = iris[feature].quantile(0.25)
    Q3 = iris[feature].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = iris[(iris[feature] < lower_bound) | (iris[feature] > uppe

    print(f"{feature}:")
    print(f"Lower Bound = {lower_bound}, Upper Bound = {upper_bound}")
    print(f"Number of Outliers = {len(outliers)}\n")
```

```
Outlier Detection using IQR:

sepal_length:
Lower Bound = 3.1499999999999986, Upper Bound = 8.350000000000001
Number of Outliers = 0

sepal_width:
Lower Bound = 2.05, Upper Bound = 4.05
Number of Outliers = 4

petal_length:
Lower Bound = -3.649999999999999, Upper Bound = 10.349999999999998
Number of Outliers = 0

petal_width:
Lower Bound = -1.95, Upper Bound = 4.05
Number of Outliers = 0
```

In [ ]: